

Analysis 1
Final Exam

Exercise 1. Course Questions:

- 1) Write the characterization of $M = \sup A$, where $A \subset \mathbb{R}$.
- 2) Recall the definition of the integer part function (floor function) of a real number x .
- 3) Recall the definition of two adjacent sequences.

Exercise 2.

- 1) Prove by induction the following assertions: $\forall n \in \mathbb{N}^*, \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2$.
- 2) Find, if they exist, the infimum and supremum of the following set:

$$A = \{ \sqrt{n+1} - \sqrt{n}, n \in \mathbb{N} \},$$

- 3) Writing $n = 4p + r$, $r = 0, 1, 2, 3$, prove that:

$$E\left(\frac{n-1}{2}\right) + E\left(\frac{n+2}{4}\right) + E\left(\frac{n+4}{4}\right) = n.$$

Exercise 3.

Let $a, b, u_0, v_0 \in \mathbb{R}$, where $u_0 < v_0$ and $0 < b < a$. For all $n \in \mathbb{N}^*$, we define

$$\begin{cases} u_{n+1} = \frac{au_n + bv_n}{a+b}, \\ v_{n+1} = \frac{av_n + bu_n}{a+b}. \end{cases}$$

1. Show that $\forall n \in \mathbb{N}, u_n < v_n$.
2. Study the monotonicity of the sequences $(u_n)_{n \in \mathbb{N}^*}$ and $(v_n)_{n \in \mathbb{N}^*}$.
3. Express $v_n - u_n$ in terms of $v_1 - u_1$. Deduce that $(u_n)_{n \in \mathbb{N}^*}$ and $(v_n)_{n \in \mathbb{N}^*}$ are adjacent.

Exercise 4.

I) Investigate the differentiability of the following function:

$$f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$

II) Consider the function f defined on $[1, +\infty)$ by:

$$f(x) = \ln(x + \sqrt{x^2 - 1})$$

- 1) Calculate $f'(x)$.
- 2) Let $x \in (1, +\infty)$.
- i. Show that there exists $c \in (1, x)$ such that:

$$\frac{\ln(x + \sqrt{x^2 - 1})}{x - 1} = \frac{1}{\sqrt{c^2 - 1}}.$$

- ii. Deduce: $\lim_{x \rightarrow 1^+} \frac{\ln(x + \sqrt{x^2 - 1})}{x - 1}$.